Tools 4.3: Collective Action, Public Goods, and Norms — Why Do Groups Fail to Get What They All Want? (AD)

Political Science — Master Notes · by Jakob V. Stangel

Required reading: Bueno de Mesquita, *Political Economy for Public Policy*, **ch. 4** · *Optional:* **Elinor Ostrom**, *Governing the Commons*, **ch. 1**. *(Lecturer: Anna Denisenko — deck "4.3 Collective Action, Public Goods, and Norms.")*

Theme of the lecture. Lecture 4.1 asked which outcomes are *good*; 4.2 asked which will *happen*. This one asks the uncomfortable follow-up: **why does the outcome that happens so often be one that nobody wants?** The answer is the **collective action problem** — the gap between **individual rationality and group rationality.** Each person does best by **free riding** (letting others pay), so groups fail to provide **public goods** and over-use **shared resources**, even when everyone would be better off cooperating. We diagnose the two kinds (**free-rider** vs **coordination**), locate them in the **goods typology** (excludable × rival), and build a **toolkit** to escape them — coercion, selective incentives, repeated interaction, and changing expectations — applied to **climate politics**.

> ★ **The heart of the lecture — individual rationality ≠ group rationality.**
> • **A collective action problem** exists when (1) an outcome would make everyone better off (a Pareto improvement), yet (2) the choices individuals make on their own don't get the group there — the group is stuck in a **Nash equilibrium that is not Pareto efficient.**
> • **Free-rider problems** come from **externalities** and live with **non-excludable goods** (public goods · common-pool resources): I benefit whether or not I contribute, so I don't — **free riding is a dominant strategy.** Bigger groups make it worse (**Olson**).
> • **Coordination problems** have **multiple equilibria**: everyone would act *if sure others would*, so everyone waits. Here nobody wants to cheat — they just need to **expect** the good outcome.
> • **The toolkit:** first **diagnose**. Free-rider → **change payoffs** (coercion · selective incentives · repeated interaction). Coordination → **change expectations** (communication · focal points · leaders · signals).

Red thread: *(1) Is this a free-rider or a coordination problem? (2) What kind of good is at stake (excludable? rival?)? (3) Which tool fits — change payoffs, or change expectations?*

Key concepts to apply: *collective action problem · individual vs group rationality · Pareto-inefficient Nash equilibrium · externality (positive/negative) · **excludable · rival** → the goods typology (private · club · common-pool · public) · free riding · dominant strategy · the public-goods game · conditional cooperators · peer punishment (Fehr & Gächter) · group size (**Olson 1965**, The Logic of Collective Action) · tragedy of the commons (**Hardin 1968**); Ostrom's challenge · coordination problem · multiple equilibria · threshold/cascade effects · focal points (Schelling) · the toolkit: **coercion · selective incentives (Olson) · repeated interaction / shadow of the future (Axelrod) · changing expectations** · social norms · conditional cooperation.*

What is a collective action problem?

Core: the central puzzle of the lecture — a group can get **stuck at an outcome nobody wants.**

A **collective action problem** exists when:

1. there is an outcome that would make **everyone in the group better off** (a Pareto improvement), **but**
2. the choices individuals make **on their own** do not lead the group there.

→ In game-theory terms (Lecture 4.2): the group lands in a **Nash equilibrium that is not Pareto efficient.** The recap is last lecture's housing game: **(Don't, Don't) = (2,2)** is the equilibrium even though **(Allow, Allow) = (3,3)** is better for both — each city hopes the other builds; each **free rides.**

Central idea for today: individual rationality ≠ group rationality. Everyone acting sensibly for themselves produces a result that's bad for all of them.

Two kinds of collective action problem

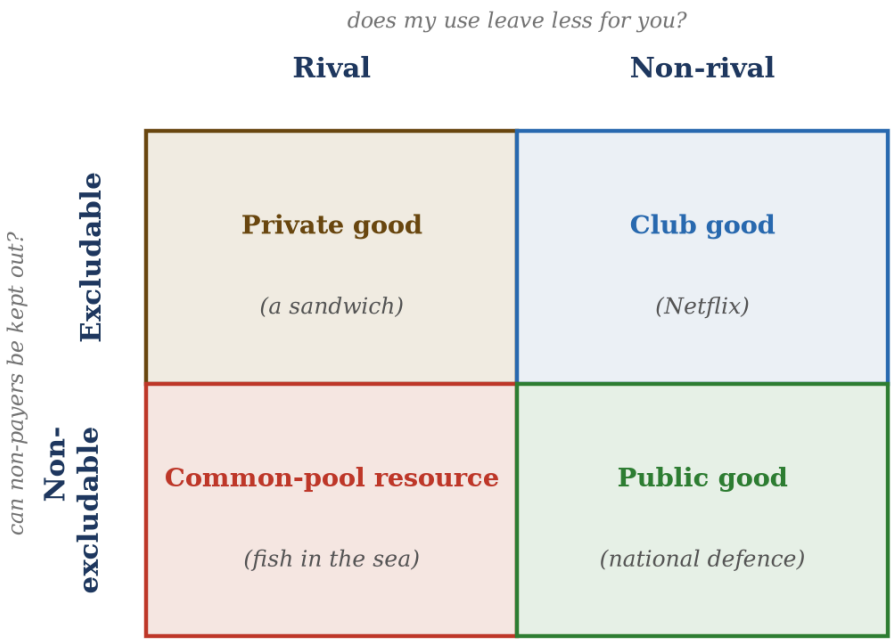
Core: diagnosis comes first — the fix depends on **which** problem you have.

	Free-rider problem	Coordination problem
Logic	**doing nothing is better for me** — even if I know others act	I'd act **only if sure others will**; so everyone waits
Result	nobody contributes (empty pot)	nobody moves (empty protest square)
Equilibrium	one bad equilibrium (a dominant strategy to defect)	**multiple** equilibria (a good one and a bad one)
Fix	**change payoffs**	**change expectations**

Externalities & the goods typology

Core: free-rider problems aren't random — they appear with **certain kinds of goods**, which you can spot with two questions.

- An **externality** is an effect of my action on others that I **don't pay for (or get paid for)**: a town's **fireworks** (positive — anyone can watch, my watching doesn't block yours); **fishing boats** sharing a sea (negative — every fish I take is one fewer for you).
- **Two questions about any good:**
  - **Excludable?** — can people who don't pay be **kept out**?
  - **Rival?** — does my use **leave less** for you?



Free-rider problems live in the bottom row — you cannot keep free riders out.

How to read it. Cross the two questions and you get **four kinds of good. Free-rider problems live in the bottom (non-excludable) row** — you **cannot keep free riders out**, so people under-provide (**public goods**) or over-use (**common-pool resources**).

★ The free-rider problem

Core: with a **public good**, I get the benefit **whether or not I contribute** — so the rational choice is **don't contribute.** That's **free riding**, and it's a **dominant strategy** (the same reasoning as "Don't" in the housing game).

> *If enough others contribute, the good is provided anyway → I save my effort. If others don't, my small contribution won't matter → I save my effort. Either way: don't contribute.*

The public-goods game (the classroom experiment): each student has 10 points; **Keep** them, or **Contribute** to a pot that is **doubled and split equally** among all. With 4 students, your 10 points become 20 in the pot but you get back only **20/4 = 5** — so **Keeping always pays 5 more**, no matter what others do. A **prisoner's dilemma with 4 players.**

But real people aren't pure free riders:

- In round 1 many **contribute a fair amount**; they're **conditional cooperators** — "I'll do my part *if* others do theirs."
- Over repeated rounds, contributions **fall** (cooperators tire of being exploited).
- Letting players **punish** free riders — at a cost to themselves — **keeps contributions high** (**Fehr & Gächter 2000**). *(This is where **norms** enter: informal punishment sustains cooperation.)*

Group size — Olson (1965)

Core: **the larger the group, the harder** collective action — the key insight of Mancur Olson's *The Logic of Collective Action*.

Scale the public-goods game from 4 to **40** members: your 10 points still become 20 in the pot, but you get back only **20/40 = 0.5.** Large groups struggle because:

1. each member's **share of the benefit is smaller**;
2. each contribution is **less noticeable** ("my effort won't matter");
3. it's **harder to monitor and pressure** free riders.

→ *Counter-intuitive upshot:* a **small** group with concentrated interests (a handful of firms) often out-organises a **large** diffuse public — which is why narrow lobbies can beat the general interest.

The tragedy of the commons

Core: the mirror image of under-provided public goods — **over-used common-pool resources.** Same free-rider logic.

- **Hardin (1968):** a pasture open to all herders. Each extra cow gives its owner the **full** benefit, but the cost (**overgrazing**) is **shared by all** → everyone adds cows until the pasture is ruined.
- **Real example — Atlantic cod** off Newfoundland: decades of overfishing (supercharged by **factory trawlers**) collapsed the stock; **Canada closed the fishery in 1992**, costing tens of thousands of jobs.
- **The symmetry: public good** → people contribute **too little**; **common-pool resource** → people take **too much.** Both are the **free-rider problem.**
- **Ostrom's challenge (optional reading).** Elinor Ostrom showed Hardin's "tragedy" is **not inevitable**: real communities often **self-govern** commons successfully — without either the **state** or **privatisation** — through local rules, monitoring and graduated sanctions. *(Ostrom ch. 1 was not in the attached files; this is the gist.)*

Coordination problems

Core: the second trap — here **nobody wants to cheat**; the problem is **expectations**, because there is **more than one equilibrium.**

The protest game. Two citizens decide whether to **Join** a protest. If only one joins, she's arrested; if both join, the dictator falls; if neither joins, he stays.

	B: Join	**B: Stay home**
A: Join	**3, 3**	0, 1
A: Stay home	1, 0	**2, 2**

- **Two equilibria:** everyone joins **(3, 3)** *or* nobody joins **(2, 2).** Both are stable.

- **The trap:** I stay home **because I fear you'll stay home.** One equilibrium is not Pareto efficient — and which one we land in depends on **what we expect.**
- **Threshold / cascade effects — Leipzig, 1989:** a few thousand East Germans joined the Monday demonstrations in September; by **23 October, 300,000+** had joined. As more protested, others became willing → a **cascade.**
- Other coordination problems: **which side of the road** to drive on (**Sweden switched overnight in 1967**), which **technical standard** firms adopt (phone chargers).

★ The toolkit — how groups escape

Core: **diagnose first**, then pick the matching tool. Free-rider problems need **changed payoffs**; coordination problems need **changed expectations.**

Problem	Tool	How it works
Free-rider	**1. Coercion**	the **state** requires/forbids an action and enforces it (police, courts, fines)
Free-rider	**2. Selective incentives** (Olson)	a **private** reward/cost given **only to contributors**
Free-rider	**3. Repeated interaction**	cooperate today to keep cooperation tomorrow (the shadow of the future)
Coordination	**4. Change expectations**	communication, focal points, leaders, signals → everyone expects the good equilibrium

1. Coercion — the state as solution. (Recall The State lecture: a classic justification for the state is solving what individuals can't alone.) **Bans** (CFCs, indoor smoking, closed fishing seasons); **mandates** (compulsory schooling, conscription); **compulsory voting** (Australia); **standards** (car emissions, building codes). *Limits:* the state needs **information & capacity**; the **liberal concern** (Lecture 4.1) that coercion limits freedom; **who controls the state** (coercion can serve organised groups); and in international politics **there is no world government** to coerce states (recall Realism: anarchy).

2. Selective incentives (Olson). A **private** benefit given **only** to contributors turns a public-good problem into a **private choice** ("join to get the benefit"). **Business associations** (Dansk Industri: legal advice, data, networks); **professional associations** (insurance, journals); **social movements** (friendship, identity). *The union puzzle:* unions win gains for **all** workers, members or not — so why do **~2/3 of Danes** pay dues? Partly **selective incentives**: in the **Ghent countries** (Denmark, Sweden, Finland, Belgium) **unemployment insurance** runs through **union-linked funds** → unionisation is far higher. *(And **norms** matter: in some workplaces, not joining is socially costly.)*

3. Repeated interaction — the shadow of the future. One-shot games invite defection; **repeated** ones support cooperation via **tit-for-tat** ("I cooperate as long as you do; if you cheat, I stop"). Works best when interaction **continues**, players **value the future**, and cheating is **observable** (**Axelrod 1984**).

4. Coordination → change expectations. Payoffs can stay the same; everyone just needs to **expect** the good outcome: **communication** (here promises are **credible**, because keeping them is in your own interest — unlike the free-rider experiment's cheap talk); **focal points** (**Schelling 1960** — an answer

that "stands out": a tradition, a date, a famous square); **leaders / first movers** (someone goes first and shows it's safe); **public signals and laws** (tell everyone what everyone else will do — **Sweden's Dagen H, 1967**; US **standard time zones, 1883**). *Once everyone expects the good outcome, it enforces itself.*

The climate case — why does Denmark cut?

Core: the running application — use the toolkit to explain the **Danish Climate Act** (cut emissions **70% by 2030**) when Denmark is just **~0.1%** of world CO_2 (China ~30%, US ~13%).

- **Diagnose:** a stable climate is a **public good** (non-excludable, non-rival); the **externality** is global; it's a **free-rider problem**; and by **Olson** it's a **hard group** (≈200 countries, each tiny).
- **Denmark's best choice** (pure free-rider logic): **pollute** — 0.1% can't change the climate. *So why does it cut?*
- **Tools: coercion** works only **inside** a state (no world government — anarchy), so between countries it's **incentives / repeated interaction** (the EU carbon border fee; **Paris** pledges-and-reviews every 5 years).
- **Change Denmark's incentives:** Danish **green industry** profits whenever Denmark cuts (+2) — turning a losing free-rider game into a winning one.
- **Verdict (three views):** free-rider logic → **no**; incentives → **yes** (green industry profits); **first mover** → Danish **wind** helped make clean energy **cheaper for everyone**, which shifts the problem **from free-rider toward coordination** (as cutting gets cheaper, more will cut). *Open question for next lecture: who should cut most — today's big emitters or yesterday's? That's a **bargaining** problem.*

Key terms

Term	Meaning
Collective action problem	a Pareto-improving outcome exists, but individual choices don't reach it
Individual vs group rationality	each acting sensibly → a result bad for all
Externality	an effect of my action on others I don't pay (or get paid) for
Excludable / rival	can non-payers be kept out? / does my use leave less for you?
Public good	non-excludable **and** non-rival (national defence) — under-provided
Common-pool resource	non-excludable but rival (fish) — over-used
Club good	excludable but non-rival (Netflix)
Free riding	enjoying a good without contributing (a dominant strategy)
Conditional cooperator	"I'll contribute if others do"
Olson's logic	larger groups find collective action harder
Tragedy of the commons	shared resource ruined by individually rational over-use (Hardin)
Coordination problem	multiple equilibria; the issue is expectations, not payoffs
Threshold/cascade	each joiner makes others willing → sudden mass participation
Selective incentive	a private reward/cost given only to contributors (Olson)
Focal point	an option that "stands out" and coordinates expectations (Schelling)
Shadow of the future	repeated interaction makes cooperation pay (Axelrod)

Exam pointers

- **Define a collective action problem** and state the slogan: **individual rationality ≠ group rationality.**
- **Classify a good** with excludable × rival and say why free-rider problems live in the **non-excludable** row.
- **Explain free riding** as a dominant strategy; use the public-goods game and **Olson's** group-size logic.
- **Contrast free-rider vs coordination** problems — and match each to its fix (**change payoffs** vs **change expectations**).
- **Run the toolkit**: coercion · selective incentives · repeated interaction · focal points — with an example of each (union dues; Ghent; Leipzig 1989; Dagen H).
- **Apply it to climate**: why Denmark "should" free-ride, and the incentive/first-mover reasons it cuts anyway.

★ Fun facts & memorable details (Tools 4.3: Collective Action)

> The sticky bits from the collective-action lecture.

- **Individual rationality ≠ group rationality.** The whole lecture in five words: everyone being sensible can leave everyone worse off.
- **The small group usually beats the big one.** Olson's twist: a handful of firms with concentrated stakes out-organise millions of diffuse citizens — which is why narrow lobbies so often win.
- **Punishment keeps cooperation alive.** In public-goods games, letting people fine free riders (at a cost to themselves) stops contributions from collapsing — the experimental root of social norms (Fehr & Gächter).
- **The cod never came back.** Factory trawlers + a shared sea = tragedy of the commons; Canada closed the fishery in 1992 and tens of thousands lost their jobs. A commons can be ruined *permanently*.
- **A revolution in six weeks.** Leipzig went from a few thousand protesters in September 1989 to 300,000+ by late October — a coordination cascade, each joiner making the next one braver.
- **Sweden changed sides overnight.** "Dagen H," 3 September 1967: the entire country swapped from driving on the left to the right at once — a coordination problem solved by a public signal everyone obeyed.
- **Denmark is 0.1% of emissions — and cuts anyway.** Pure free-rider logic says don't bother; it cuts because green industry profits and because going first made clean energy cheaper for everyone — nudging the world from free-riding toward coordination.
